Campus City Emergency Supply Distribution

B.Tech Minor – Micro Project Report

Submitted by

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Abstract

Emergency supply distribution is an important problem in logistics planning, especially in campus cities where essential resources must reach facilities quickly and efficiently. This project focuses on selecting optimal warehouses and allocating supplies to facilities while minimizing operational and transportation costs. A Mixed Integer Linear Programming (MILP) model is used to determine the most efficient solution under budget and capacity constraints. Visualization tools such as Folium are also used to display the geographical distribution of warehouses and facilities.

1 Problem Statement

This project focuses on managing emergency supply distribution in a campus city. In this system, several warehouses are available to store and distribute emergency resources to important facilities such as hospitals, dormitories, academic buildings, and public centers. However, operating all the warehouses is not economically efficient because each warehouse has high construction, operational, and transportation costs. At the same time, different facilities require different amounts of supplies, and the system must work within a limited budget and capacity constraints. Therefore, the main goal of this project is to identify the best set of warehouses to operate and determine how supplies should be distributed from these warehouses to different facilities. The objective is to ensure that all facility demands are satisfied while minimizing the total annual cost of operation and transportation.

2 Mathematical Formulation

To resolve the location-allocation problem, a Mixed-Integer Linear Programming (MILP) model was developed.

2.1 Decision Variables

- $y_w \in \{0, 1\}$: Binary variable; 1 if warehouse w is selected for construction/operation, 0 otherwise.
- $x_{w,f} \geq 0$: Continuous variable representing the quantity of units shipped from warehouse w to facility f per day.

2.2 Objective Function

The objective of the model is to minimize the total annual cost of operating warehouses and transporting supplies.

$$Z = \sum_{w \in W} \left(\frac{Const_w}{10} + (Op_w \times 365) \right) y_w + \sum_{w \in W} \sum_{f \in F} (Trans_{w,f} \times x_{w,f} \times 365) \tag{1}$$

where,

- $Const_w$ = Construction cost of warehouse w
- Op_w = Daily operational cost of warehouse w
- $Trans_{w,f}$ = Unit transportation cost from warehouse w to facility f
- y_w = Binary variable indicating whether warehouse w is selected
- $x_{w,f}$ = Quantity transported from warehouse w to facility f

2.3 Constraints

The optimization model is subject to the following constraints:

1. Redundancy Requirement

$$\sum_{w \in W} y_w = 2 \tag{2}$$

Exactly two warehouses must be active.

2. Demand Fulfillment

$$\sum_{w \in W} x_{w,f} = Demand_f \quad \forall f \in F \tag{3}$$

All facility demands must be satisfied.

3. Capacity Limitation

$$\sum_{f \in F} x_{w,f} \leq Cap_w \cdot y_w \quad \forall w \in W \tag{4}$$

Shipments from each warehouse must not exceed its capacity.

4. Financial Constraint

$$Z \leq 1{,}500{,}000 Thetotalcostmustremainwithinthe\$1.5millionbudget. \tag{5}$$

The first term represents the annualized construction cost (10-year amortization) and fixed operational costs of selected warehouses. The second term represents the variable transportation costs based on daily shipment volumes.

To solve this problem, a mathematical optimization model called Mixed Integer Linear Programming (MILP) was used. In this formulation, decision variables represent whether

a warehouse should be opened or not and how much supply should be transported from each warehouse to each facility. The objective of the model is to minimize the total annual cost involved in operating the distribution system. This cost includes the construction cost of warehouses, their daily operational cost, and the transportation cost required to deliver supplies to facilities. The construction cost is calculated using a ten-year amortization approach, and operational costs are considered on a daily basis. In addition, transportation costs depend on the quantity of supplies delivered to each facility. By combining these factors, the model determines the most cost-efficient solution.

3 Methodology

The problem was solved using a Mixed Integer Linear Programming approach. In this method, the objective function and all constraints are expressed as linear equations. Warehouse selection is represented as a binary decision variable, meaning a warehouse can either be selected or not selected. The transportation quantities are represented as continuous variables because the amount of supply sent from warehouses to facilities can vary. One major advantage of this method is that MILP guarantees a globally optimal solution when solved using appropriate optimization algorithms.

In this project, optimization solvers such as PuLP, CBC, or Gurobi were used to compute the best solution efficiently. The model was implemented programmatically where input data such as warehouse details, facility demands, and transportation costs were read from CSV files. After defining the decision variables, the objective function and constraints were applied, and finally the solver computed the optimal solution.

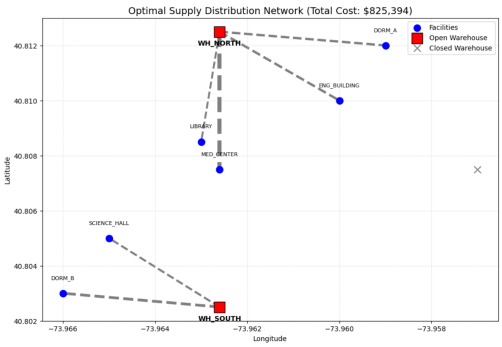
In addition to optimization, the **Folium Python library** was used to visualize the locations of warehouses and facilities on an interactive map. This helped in clearly displaying the distribution network and understanding how supplies are allocated from selected warehouses to different facilities geographically.

4 Results and Output Analysis

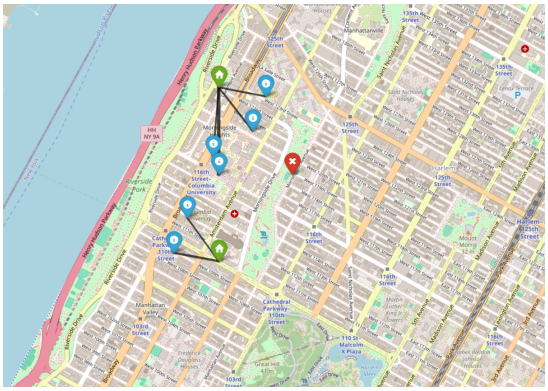
After running the optimization model, the results showed that selecting two warehouses, namely WH_NORTH and WH_SOUTH, provides the best solution for the system. These warehouses together have sufficient capacity to satisfy the demand of all facilities while maintaining lower operational and fixed costs compared to other possible combinations.

The total annual cost of operating the system was calculated to be $959,466.05. Out of this amount, $605,500 represents fixed costs such as construction and operations, while $353,966.05 represents transportation costs. Even after covering all expenses, the solution remains within the allocated annual budget, leaving a remaining budget of $540,533.95.

The model also ensured that the total daily demand of 270 units was completely satisfied. Important facilities such as hospitals and community centers received adequate supplies, and no warehouse exceeded its maximum capacity.



(a) Warehouse locations network



(b) warehouse location map

Figure 1: Visualization of the supply distribution network

5 Conclusion

This project demonstrates how optimization techniques can be used to solve real-world logistics and supply distribution problems. By modeling the campus emergency supply distribution system using Mixed Integer Linear Programming, the project successfully identified the optimal warehouses and determined the most efficient supply allocation strategy. The solution ensures that all facility demands are satisfied while minimizing overall costs and staying within budget limits.

The addition of visualization using Folium also helps in clearly understanding the geographical distribution of warehouses and facilities. This approach provides a reliable and scalable framework that can be used for emergency planning and future expansion of the distribution system.

6 References

1. Siju Swamy Research Website: `https://sijuswamyresearch.github.io`

2. PuLP Documentation: `https://coin-or.github.io/pulp/`